Beyond the Heatmap: Toward Metaphorical Visualization of Embodied Carbon in Architectural Design

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Abstract. This study conducts a Systematic Literature Review (SLR) to formulate an exploratory exercise on alternative visual strategies to make objective data tangible for non-experts. We establish a foundation for metaphorical visualization (MET-VIS) for higher cognitive fluency and support evidence-based design in architecture. Our approach couples environmental impact data of buildings with Artificial Intelligence (AI) as an exploratory medium moving beyond compelling imagery. Our two-stage protocol involves: 1) documenting conventional against metaphorical techniques, and 2) AI-mediated MET-VIS prototyping. Metaphorical choices (deformations, textures, colors) activated non-visual cognitive registers (burden, spread, discomfort) rather than direct numerical translation. We propose a visual interaction model fostering accessible built environmental awareness, with user testing as a next step to validate cognitive fluency.

Keywords: 3D Metaphors, Environmental Data, Building Information Visualization, Architectural Representation, AI-Mediated Design

1 Introduction

Building decarbonization urgency grows, yet digital representation of environmental data remains restricted to experts, making graphic techniques discussion inevitable. While Artificial Intelligence (AI) generates compelling architectural imagery, it risks producing passive content without active human

agency (Crawford, 2021; Guo *et. al.,* 2023; Moussaoui, 2025; Negris de Souza *et al.*, 2025; Sharma *et al.*, 2025). To surpass abstract data depiction barriers, we critique ocularcentrism and advocate multisensory design (Dunne, Raby, 2013; Pallasmaa, 2005; Zumthor, 2006). Building on Hollberg et al. (2021), this study explores metaphorical visualization (MET-VIS) to bridge data and meaning, promoting cognitive fluency through multisensory associations to support evidence-based design (Bludau *et al.*, 2025; Isaev *et al.*, 2024; Li *et al.*, 2016; Wijayawardena *et al.* (2023). Generative AI serves as an exploratory medium triggering affective associations, making objective data tangible for non-experts (those without pre-existing contextual or technical knowledge), beyond purely artistic purposes (Ponzio *et al.,* 2023). We investigate this communication issue using a two-phase methodology: a Systematic Literature Review (SLR), and AI-mediated MET-VIS prototyping. to make objective data tangible for non-expert (built environment value chain stakeholders).

1.1 Information-based building representation for LCA and MET-VIS

Two concepts served as the backbone of our approach: building environmental impact data and data visualization. The former led us to pinpoint embodied carbon (EC) as a specific indicator within Life Cycle Assessment (LCA) methodology (EN 15978:2026, European Committee for Standardization [CEN], 2026); the latter directed us towards MET-VIS techniques. Visualizing EC in architectural components exemplifies the role of material emission adjustments for climate impact reduction (Adams et al., 2019; Dixit *et al.*, 2024). Although operational carbon (OC) dominates building emissions, it is most readily minimized (Röck *et al.*, 2023); yet mitigating OC through energy efficiency solutions often increases EC. This performance trade-off is hindered by retrospective LCA integration, opposing to early-stage inclusion (Jusselme *et al.*, 2018).

Without accessible visual strategies, non-experts face high cognitive loads when interpreting spatial data (Apellániz *et al.* 2024; Forth *et al.*, 2023). In resonance with Cerdas et al. (2017), Gadelhak et al. (2017), and Wijayawardena *et al.* (2023), our interest lies in graphic experimentation that questions visual syntaxes for improved comprehension in the architecture, building, and construction field. This contrasts with the different types of visualization clustered by Hollberg et al. (2021) (pie/bar charts, heatmaps, bubble maps, virtual reality, etc.), interpreted here as conventional visualization methods.

We prioritize MET-VIS's representational efficacy over numerical accuracy alone, which, according to Vera-Piazzin *et al.* (2023), depends on clarity,

precision, and efficiency, considering that visual information should be both useful and meaningful. Different design environments (Building Information Modeling - BIM, parametric tools, generative AI) possess unique graphic syntaxes. Leveraging AI might help highlight perceptual data categories (distribution, relationship, weight, composition), rendering abstract metrics tangible without requiring expert knowledge.

2 Methodology

Our quantitative-qualitative framework covers two phases: (1) an SLR surveying conventional building data visualization against metaphorical techniques; and (2) AI-mediated MET-VIS exploratory prototyping. Three assessment criteria guide the evaluation: (a) limitations of current methods towards the MET-VIS approach; (b) translation of abstract data into tangible visual features; (c) assessment of Generative AI as an interlocutor for MET-VIS.

2.1 Phase 1: Systematic Literature Review Protocol

The SLR underpins graphic structures for digital metaphorical diagrams designed aiding non-experts' LCA comprehension.

Protocol: The PICOC framework breaks down objectives (*Population, Intervention, Comparison, Outcomes,* and *Context)* (Carrera-Rivera *et. al., 2022)* targeting non-experts, LCA data, design processess, visualization methods, and MET-VIS.
Research Questions: *What class and type of limitations are documented regarding data visualization, and what principles are explored to address those limitations?; What existing precedents of metaphorical or non-abstract data representation exist in architecture, graphic design, or other discipline found can inform and substantiate possible visual directions for LCA representation?; Which graphic limitations and principles were discussed in terms of altering the intended target audience that has access to the communicated data?*
Database, Search Strategy, and Inclusion/Exclusion Criteria: SciELO, Redalyc, and Scopus (2016–2026, Open Access, English/Portuguese/Spanish) were searched across two tracks: "General Approach" (multidisciplinary metaphors) and "Specific Approach" (architectural applications) (Figure 1, top). For each track, a specific group of strings was determined.

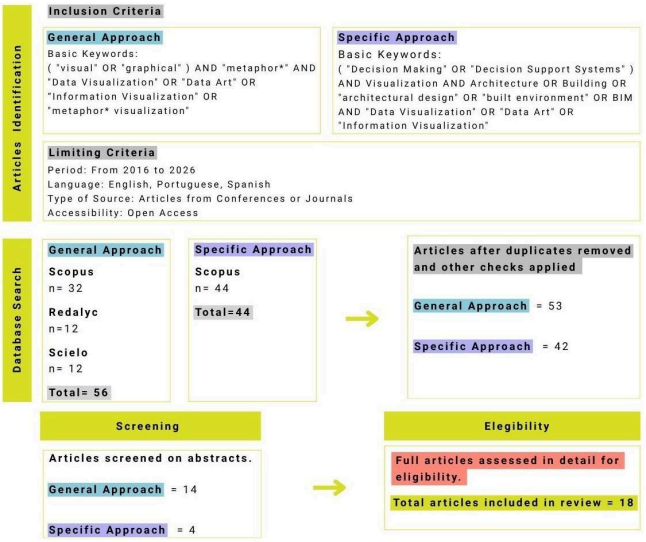


Figure 1. SLR Process and Results. Source: Authors (2026)

Screening, Selection Criteria, and Quality Assessment (QA): A 0–4 Likert checklist evaluated candidate papers, requiring scores of 3–4 across criteria for inclusion (Figure 1, bottom):

- *Relevance:* Does it have a strict relationship with the fields of architecture or graphic design? [context and population]
- *Rigor:* Does it have a clear goal statement based on understanding/facilitating/improving the impact of visualization beyond conventional ones? [intervention]
- *Credibility:* Is the method applied tailored to assess specific aspects of visual representation? [comparison]
- *Reporting:* Are there conclusions concerning the characteristics of the visual representations considered in the paper's context? [outcomes]

Data Extraction (DE): Data were extracted and classified by:

- *Research field:* Architecture, graphic design, or data representation;
- *Communication intention of visual graphics:* Purpose, data/framework type, and audience;
- *Limitations:* Documented visualization issues;
- *Evaluation principles of initial examples:* Design principles guiding the initial visual analysis
- *Findings on visualization:* Selected visual elements and new representation modes.

Google AI Studio (Gemini 3.6 Flash) served as an auxiliary tool using system instructions grounded in our SLR protocol, organizing the researcher's preliminary synthesis, which the authors interpreted to summarize and formulate propositions for the primary visual metaphors.

Summarizing Results: DE findings were analysed via a semantic-formal approach, classifying visual elements through representational considerations from Ching (2008) and Dondis (2003), and Souza (2023) into "Basic Elements" (point, volume, texture, etc.) and "Combination Forms" (balance, dimension, subtraction, addition, etc.). Finally, technical LCA concepts and definitions (e.g., reference service life, scope, boundaries), specifically for EC, were integrated for semantic precision (CEN, 2026) to construct the graphic structure framework.

2.2 Phase 2: AI-Mediated Exploratory Prototyping and Translation

To address traditional data representation challenges, an AI co-creative partner was integrated based on Phase 1 results.

Functional Requirements: Tool selection required multimodal input capacity to process text and images simultaneously, prioritizing creative interpretation over photorealistic fidelity. A reference image setup and in-process hand-drawing tools were also considered to preserve human agency in future study stages.

Prompting Strategy: Textual inputs were structured around specific constraints (nouns, verbs, adverbs, adjectives) derived from semantic analysis of the WorldGBC (2019) report, grounding AI's outputs in technical EC meanings. Attention control (textual highlights) and structural delimiters (special markers and symbols) were applied alongside visual boundaries regulated via EC descriptions and spatial syntax guidelines, ensuring metaphorical translations adhered to Phase 1 graphic rules (Li *et al.*, 2026). A generic 3D interior served as a structural blank space to control primary architectural components (walls, slabs, windows, doors), followed by a recursive generation loop for refinement.

Evaluation Criteria for Generated Prototypes: Generated outcomes were evaluated prioritizing open observational assessment over a rigid checklist, subjectively correlating the AI outputs against intentional visual descriptions grounded in visual elements. This qualitative filtering verified how harmoniously the visual results integrated the established rules, baseline structural hierarchies, and the overarching MET-VIS conceptual framework.

3 Results

Phases 1 and 2 outcomes are synthesized in Figure 2 with additional material accessible via GitHub (https://github.com/larissanegris/sigradi2026_Docs_LNS-SOC).

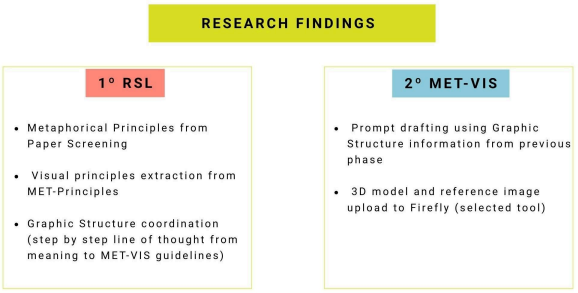


Figure 2. Results summary sequence - SLR and MET-VIS. Source: Authors (2026)

3.1 Phase 1: SLR Findings and Proposed Graphic Structures

The sequential steps of SLR-DE led to compiled results to propose the graphic structure as follows.

SLR-DE: A comprehensive review identified commonalities in alternative visual strategies for complex data visualization, particularly regarding the cognitive limitations of conventional methods (static symbols, high levels of abstraction, and 2D multidimensional representations that translate data without communicating their actual impact). Paper assessment was articulated with the AI Studio assistant supported the metaphorical principles' identification. Those highlighted in Figure 3 were selected for their clarity in leading visual principles (basic elements and composition forms) that together could support our MET-VIS framing.

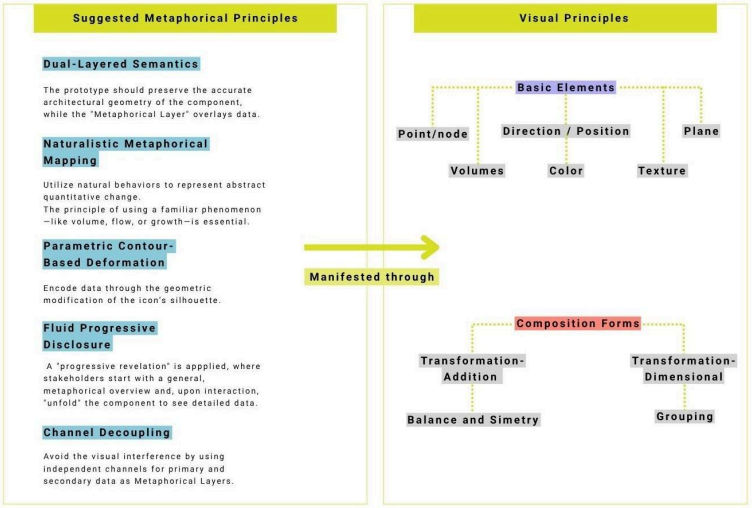


Figure 3. SLR-DE findings. Source: Authors (2026)

Graphic structure: Beginning with a generic EC definition, we identified the main semantic attributes to explain this indicator. The Vocabulary Matrix partitions EC into tangible, text-based directions, solidifying abstract concepts into more approachable ideas. A Composite Metaphor emerged from connecting the Vocabulary Matrix and the SLR-DE findings (Figure 4), encompassing two superimposed metaphor types: Biomorphic Metaphor for carbon data and Metaphorical Worlds for architectural setup. Derived Visual Principles provided the graphic-based, tangible language best suited to the metaphors, arriving at the signifier of the graphic structure: the AI-mediated MET-VIS exploratory images. Figure 4 depicts the synthesis for our experimental perspective. Details for each step are provided:

1. Signified: EC defined as a high-intensity, carbon-intensive lifecycle challenge.
2. Semantic: Abstracted EC information into text-based directions (e.g., "massive," "spread," "mitigate").
3. Composite Metaphor: Superimposed "Biomorphic Metaphors" (carbon as ‘parasitic’ organism) onto "Metaphorical Worlds" (architectural components as containers).
4. Visual principles: Applied volumetric scaling (size = intensity), boundary alignment (graphic = structural limits), and visual "burden" (distortion = high impact).
5. Signifiers: Generated prototypes via multimodal workflows, combining geometric sketches with the metaphorical prompt logic.

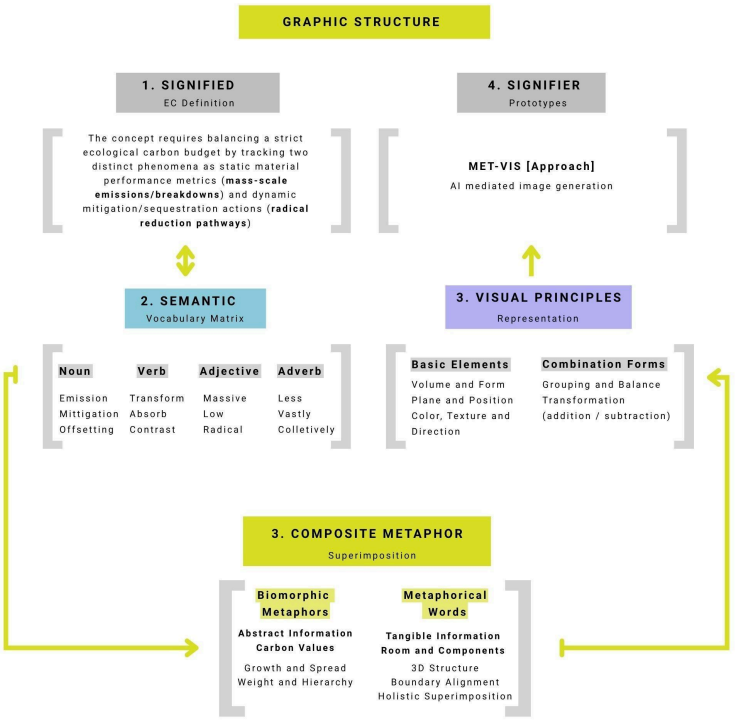


Figure 4. Graphic Structure. Source: Authors (2026)

3.2 Phase 2: MET-VIS Exploratory Prototypes

Concepts were depicted using a color-coded 3D baseline layer (Figure 5, right): yellow slabs, pink walls, blue windows, and green doors. The process followed a recursive loop: (i) prompt drafting via Gemini 3.1 Flash Lite to encode MET-VIS requirements (Figure 5, left); (ii) 3D model and reference image upload to Firefly (selected tool) employing 'Gemini 3.1 w / Nano Banana 2' model configuration; (iii) iterative refinement.

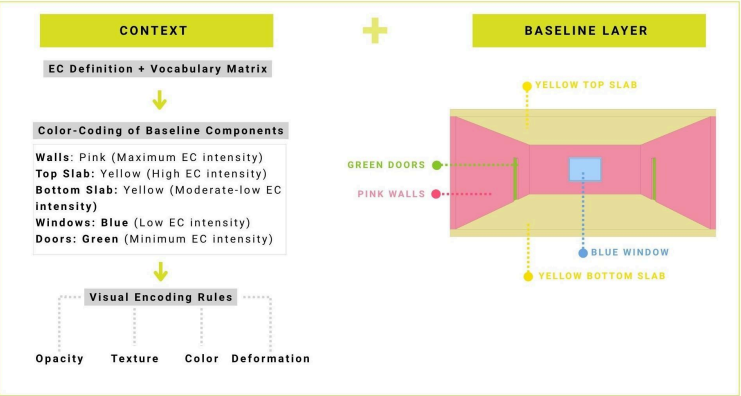


Figure 5. Prompting and baseline layer conceptualization. Source: Authors (2026)

Prompting: Prompt drafting was refined in Gemini to resolve structural inconsistencies (baseline layer discrepancies and visual element misalignments).

Visualization: Image generation was constrained by four parameters to anchor the metaphorical translation (Figure 6), which were described during the prompting step:

1) *Opacity:* Semi-transparent volumes highlighting component edges to emphasize the data layer over the geometry.
2) *Texture:* Parasitic carbon-chain patterns applied internally to evoke the "Organism" metaphor.
3) *Color:* Supporting blue-to-red gradients representing EC intensity.
4) *Deformation:* Biomorphic spikes protruding from boundaries to haptically signal high-impact zones.

To test comprehension of EC behavior, we established a component-based Intensity Hierarchy: Walls > Top Slab > Bottom Slab > Window > Doors. This hierarchy was defined as a preliminary ordering, as the current focus remains on visual outcomes; benchmark EC values and precise quantitative testing are reserved for future stages.

Restrictive *versus*. Latent Generation: The initial set of tests, labeled "restrictive layer-based" (Figure 6, middle), utilized the full suite of rules, requiring iterations to align the graphic outcomes with the baseline hierarchy. Conversely, "latent-based" tests (Figure 6, bottom) emerged from the need to explore more open conditions: prompts retaining metaphorical definitions but with less controlled visual rules, producing a distinct MET-VIS concept. However, without pre-defined Intensity Hierarchy, AI failed to align visual outputs with intended proposals, demonstrating that structural hierarchy must be established at the prompt stage

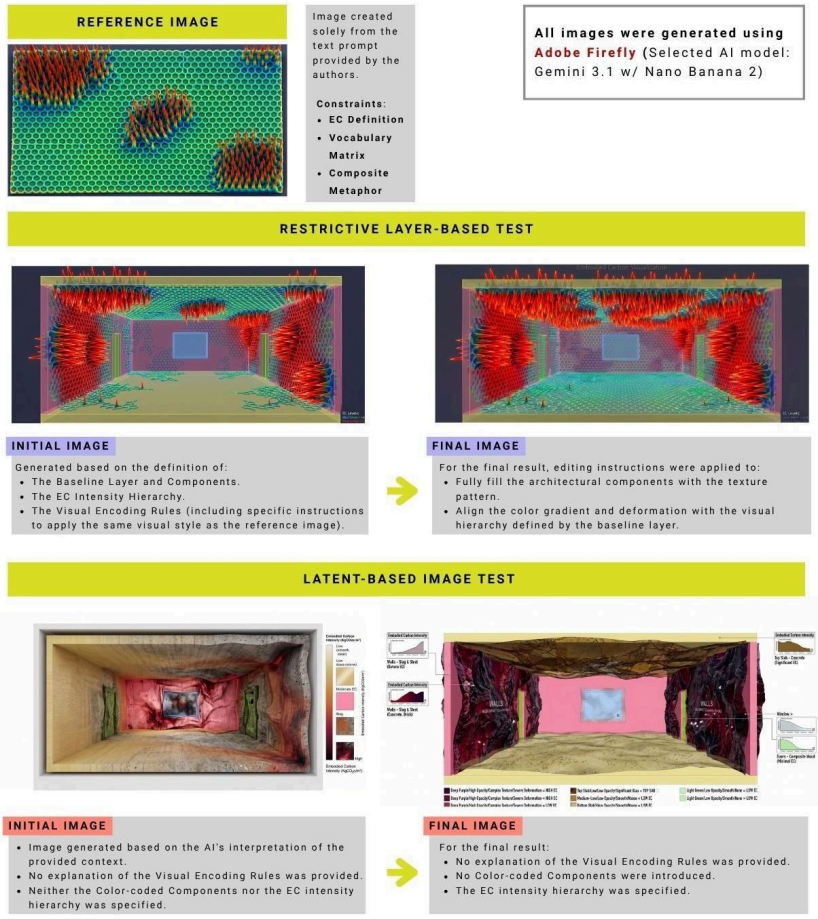


Figure 6. MET-VIS Prototypes: Primary results. Source: Authors (2026)

4 Discussion

From the SLR up to the MET-VIS propositions, we formulated three evaluation discussion fronts that follow.

4.1 Limitations of Current Methods towards the MET-VIS Approach

The SLR confirmed that human cognitive processing limits hinder interpreting complex datasets, whereas 3D visualizations enhance cognitive fluency for heterogeneous data. The primary limitation was "passive representation"

(data displayed to be read rather than explored), characterized by weak formalization, high abstraction, and incompleteness. Traditional 2D visualizations often fail to capture the multidimensional, irregular nature of increasingly voluminous datasets, leading to high-effort cognitive processing and incomplete interpretation. This disconnection between current visualizations and intuitive human perception results in a persistent gap between data and meaning, compromising comprehension and engagement for non-expert stakeholders. The literature thus advocates for the democratization of data representation, which is precisely what led us to translate abstract data into tangible visual features for our MET-VIS approach.

4.2 Translation of Abstract Data into Tangible Visual Features

The literature review confirmed that while metaphorical representation exists across various fields, a democratic visual strategy for LCA data in architecture is currently absent. Most existing non-conventional visualizations fail to achieve the level of abstraction and intuitive association required for accessible non-expert understanding. Our MET-VIS framework addresses this gap by identifying visual metaphorical principles, manifested through foundational 'basic elements' and 'combination forms', allowing complex LCA metrics to be translated into a graphic syntax that prioritizes cognitive fluency over numerical display. Its claim as a novel technique for communicating EC impact, however, should be tempered until non-expert perception evaluation.

4.3 Assessment of Generative AI as an Interlocutor for MET-VIS

Digital representations in building design recurrently couple objective and visual data. Building on our critique of direct numerical translation into static symbols and high abstraction, we positioned AI as a useful tool for visual exploration, turning abstract concepts into tangible features through a visual syntax in which symbology operates both graphically and textually. The MET-VIS approach further redirected attention to the multisensory mental models we intended when critiquing ocularcentric representation: metaphorical choices activate non-visual cognitive registers (e.g., deformations, textures, colors, etc., evoke a 'burden' or 'spread' sense). Effective outcomes transcend aesthetic appeal. Overcoming decorative outputs requires conscious prompting: specific constraints, attention control operators, and structural delimiters, alongside visual boundaries regulation as a starting point for exploration. These elements proved critical when manipulating the multimodal capacity of such technologies, and none worked in isolation. Conversely, where AI’s latent capacity could lead to more

innovative and open solutions, we deliberately relied on it to explore alternative combinatory paths. However, image generation in this scenario can be time-consuming, if not in terms of rendering efforts, in terms of the recursive loop required to adjust both prompt and graphical outcomes.

5 Conclusion

The exploratory visual approaches outlined here offer preliminary solutions toward a more intuitive means of interpreting carbon accounting data, with full potential yet to be demonstrated. Two key contributions emerged from this work. First, a novel guideline within the debate on the limits of conventional LCA visualization methods, opening an in-depth discussion on MET-VIS content. Second, a deliberate framework for handling AI tools: one that calibrates when to rely on open guidelines to explore latency and when to call for objective, concise definition in image generation. Ultimately, the value of this phase lies in structuring the questions and visual affordances and constraints that will guide subsequent research phases, directed at consolidating MET-VIS as a framework for non-experts, with user testing as a critical future step.

Declaration of AI use. AI was used for language editing and text reduction; technical usage is detailed in Methodology.

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